Equation sheet: governing relations of the $CaCO_3$-barrier screen

P0-W3.2, contract v2. Symbolic only; every symbol is in the table overleaf and in `theory/symbols.yaml`; units in brackets.

	R-slurry: aqueous, alkaline, room temperature; casting and drying on native Al_2O_3/AlOOH	R-cell: non-aqueous, fluorine-free electrolyte, collector at anodic potential vs Li^+/Li
Links that apply	C1 adhesion (1), C2 sequestration (2), supersaturation (3), C3 nucleation and Young (4, 5), template action (6), C4 morphology (7), C5 aqueous Pourbaix (8a)	C5 Li-open stability window (8b), C4 insulating-film bound (9). Not defined here: (1)–(7), since there is no hydrated Ca^{2+} reference and no aqueous Ω

1 Adhesion in water (C1, R-slurry)

The polymer never meets a dry oxide: every contact first displaces interfacial water, so the decisive work separates the contact into two water-wetted *surfaces.*

$$W_{\mathrm{adh,wet}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}] \tag{1}$$
$$= W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}$$
$$W_{\mathrm{adh,dry}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]$$
$$W_{\mathrm{adh,wet}} = \Delta F_{\mathrm{PMF}}/A_{\mathrm{c}} \quad [\mathrm{J\,m^{-2}}] \tag{2}$$

Anchored against water iff $W_{\mathrm{adh,wet}} > 0$; $W_{\mathrm{ow}}, W_{\mathrm{pw}}$ are the dry works of the oxide and polymer with water.

2 Sequestration (C2, R-slurry)

The template raises local Ca^{2+} only if it holds Ca^{2+} more strongly than water does; the cycle starts from hydrated Ca^{2+} at the regime activity.

Cycle: Ca^{2+}(aq, bulk, $a_{Ca^{2+}}$) $\rightarrow$ Ca^{2+}(binding layer). Solution side: molal standard state at the regime ionic strength; layer side: site-volume standard state, $\Delta G_{\mathrm{site}} = -RT\ln\big[V_{\mathrm{site}}^{-1}\int_{\mathrm{site}} \mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}\mathrm{d}^3r\big]$, with w the per-ion PMF ($w \rightarrow 0$ in bulk). With one coordinating group that must be deprotonated:

$$\Delta G_{\mathrm{seq}} = \Delta G_{\mathrm{site}} + RT\ln\big(1 + 10^{\,\mathrm{p}K_{\mathrm{a}} - \mathrm{pH}}\big) \quad [\mathrm{J\,mol^{-1}}] \tag{3}$$

$$\mathcal{E} \equiv \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}} = \exp\Big(-\frac{\Delta G_{\mathrm{seq}}}{RT}\Big) \quad [1] \tag{4}$$

Valid for dilute Ca^{2+} with unsaturated sites ($\mathcal{E}c_{\mathrm{bulk}} \ll c_{\mathrm{max}}$). $\mathcal{E}$ is a *concentration* ratio: at equilibrium the Ca^{2+} activity in the layer equals the bulk one, and the enrichment sits in a local activity coefficient. Langmuir closure, with x the occupancy odds:

$$c_{\mathrm{local}} = \frac{\mathcal{E}c_{\mathrm{bulk}}}{1+x} = c_{\mathrm{max}}\Theta, \quad x = \frac{\mathcal{E}c_{\mathrm{bulk}}}{c_{\mathrm{max}}}, \quad \Theta = \frac{x}{1+x} \tag{5}$$
$$c_{\mathrm{max}} = \Gamma_{\mathrm{s}}/\delta_{\mathrm{s}} \ \ [\mathrm{mol\,m^{-3}}], \qquad \Gamma_{\mathrm{Ca}} = \Gamma_{\mathrm{s}}\Theta - \delta_{\mathrm{s}}c_{\mathrm{bulk}} \ \ [\mathrm{mol\,m^{-2}}]$$

Gibbs adsorption: binding lowers the template–liquid energy, $\mathrm{d}\gamma_{\mathrm{sl}} = -\Gamma_{\mathrm{Ca}}\,\mathrm{d}\mu_{\mathrm{Ca}}$, integrated from zero to the regime activity:

$$\gamma_{\mathrm{sl}} = \gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}} \quad [\mathrm{J\,m^{-2}}] \tag{6}$$
$$\Delta\gamma_{\mathrm{ads}} = -RT\big(\Gamma_{\mathrm{s}}\ln(1+x) - \delta_{\mathrm{s}}c_{\mathrm{bulk}}\big)$$
$$\partial\Delta\gamma_{\mathrm{ads}}/\partial\Delta G_{\mathrm{seq}} = \Gamma_{\mathrm{s}}\Theta > 0$$

$\Delta\gamma_{\mathrm{ads}} \equiv 0$ if γ_{sl} is computed at the regime activities. The bound Ca^{2+} belongs to the template–liquid interface only; γ_{sn} (template coordinating lattice ions) does not depend on $a_{Ca^{2+}}$.

3 Supersaturation (one definition)

Crystallisation is driven by the excess chemical potential of the dissolved ion pair over the solid of the registered framing (ACC or calcite).

$$\Omega = \frac{\mathrm{IAP}}{K_{\mathrm{sp}}} = \frac{a_{Ca^{2+}}\,a_{CO_3^{2-}}}{K_{\mathrm{sp}}}, \qquad \Delta\mu = k_{\mathrm{B}}T\ln\Omega \quad [\mathrm{J}] \tag{7}$$

per $CaCO_3$ formula unit (V_{m} also per formula unit); nucleation requires $\Omega > 1$. **At the template $\Omega = \Omega_{\mathrm{bulk}}$.** At equilibrium $\tilde{\mu}_{\mathrm{Ca}}$ and $\tilde{\mu}_{CO_3}$ are uniform; a mean potential ψ gives factors $\mathrm{e}^{\mp 2e\psi/k_{\mathrm{B}}T}$ that cancel in the neutral solid, and specific binding lowers the local standard potential by exactly what it raises the concentration. Neither changes $\Delta\mu$:

$$\Delta\mu = \tilde{\mu}_{\mathrm{Ca}} + \tilde{\mu}_{CO_3} - \mu_{CaCO_3} = k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}} \quad [\mathrm{J}] \tag{8}$$

The local-ideal form $\Omega_{\mathrm{li}} = (c_{\mathrm{local}}/c_{\mathrm{bulk}})\Omega_{\mathrm{bulk}}$, $\Delta\mu_{\mathrm{li}} = \Delta\mu_{\mathrm{bulk}} - \Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$, credits the binding free energy to the driving force; it is kept only as the named double-counting demonstration.

4 Classical nucleation (C3, R-slurry)

A cluster pays surface free energy before its volume gains $\Delta\mu$; the barrier is the top of that trade-off. On a substrate the cluster is a spherical cap with the same curvature.

$$\Delta G_{\mathrm{hom}}^{*} = \frac{16\pi\,\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\,\Delta\mu^2}, \quad r^{*} = \frac{2\gamma_{\mathrm{nl}}v_{\mathrm{m}}}{\Delta\mu}, \quad v_{\mathrm{m}} = \frac{V_{\mathrm{m}}}{N_{\mathrm{A}}} \quad [\mathrm{J};\ \mathrm{m}] \tag{9}$$

$$\Delta G_{\mathrm{het}}^{*} = f(\theta)\,\Delta G_{\mathrm{hom}}^{*}, \qquad f(\theta) = \tfrac{1}{4}(2+\cos\theta)(1-\cos\theta)^2 \tag{10}$$

$f(0) = 0$, $f(\pi/2) = \frac{1}{2}$, $f(\pi) = 1$; $f'(\theta) = \frac{3}{4}\sin^3\theta \geq 0$, so $0 \leq \Delta G_{\mathrm{het}}^{*} \leq \Delta G_{\mathrm{hom}}^{*}$ on $[0,\pi]$. $\Delta G_{\mathrm{hom}}^{*} \rightarrow \infty$ as $\Omega \rightarrow 1^{+}$, and $\Delta G_{\mathrm{hom}}^{*} \rightarrow 0$ as $\Omega \rightarrow \infty$.

5 Young, with one sign convention

θ is measured through the nucleus. Forming the contact removes substrate–liquid interface and creates substrate–nucleus interface over the same disc; $\cos\theta$ is that energy gain per unit nucleus–liquid energy.

$$\cos\theta = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}} \quad [1;\ \mathrm{J\,m^{-2}}] \tag{11}$$

$\theta = 0$: the nucleus wets the template completely (no barrier); $\theta = \pi$: no wetting (homogeneous barrier). $\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)}$ is the free energy per unit contact area to displace the template-bound water layer; $\Delta\gamma_{\mathrm{w}} > 0$ **is a cost** and raises θ, f and $\Delta G_{\mathrm{het}}^{*}$ ($\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = -1/\gamma_{\mathrm{nl}} < 0$). If γ_{sn} is supplied as a full in-water interfacial free energy, $\Delta\gamma_{\mathrm{w}} \equiv 0$ and is not added again. $\gamma_{\mathrm{eff}} \equiv \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ (deck form) is **not** used: Young and $\Delta G_{\mathrm{hom}}^{*}$ take the bare γ_{nl}. Reason: γ_{nl} multiplies the cap's free surface, which never touches the template, and $\Delta G_{\mathrm{het}}^{*} = f\,\Delta G_{\mathrm{hom}}^{*}$ follows only if the same γ_{nl} sits in Young and in $\Delta G_{\mathrm{hom}}^{*}$. With γ_{eff} in the denominator, $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = (\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)})/\gamma_{\mathrm{eff}}^2 > 0$ on the wetting side: a cost would lower the barrier.

6 Where the template acts (screen rule)

In the grand potential at the bulk chemical potentials every template–ion interaction sits in the interfacial energies, so the template acts only through θ; the driving force is the bulk one.

$$\Delta G_{\mathrm{het}}^{*} = f\big(\theta[\gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}},\, \gamma_{\mathrm{sn}}^{(0)},\, \Delta\gamma_{\mathrm{w}}]\big)\,\frac{16\pi\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\big(k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}\big)^2} \tag{12}$$

Rule: ΔG_{seq} enters once, through $\Delta\gamma_{\mathrm{ads}}$ in γ_{sl}; $\partial\Omega/\partial$(template inputs) $= 0$. If γ_{sl} is computed at the regime activities, add nothing. At fixed $\gamma_{\mathrm{sl}}^{(0)}, \gamma_{\mathrm{sn}}$, stronger binding *lowers* $\cos\theta$ and raises $\Delta G_{\mathrm{het}}^{*}$ (SF2): binding helps only if γ_{sn} falls by more than γ_{sl}. Lowering γ_{nl} lowers f only if $\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}$ (SF3). Proof: derivation, Section 7.

6a Accessibility of the barrier (C3)

Nuclei must form at the stated areal density within the time the template spends in the slurry and drying.

$$J = J_0\mathrm{e}^{-\Delta G_{\mathrm{het}}^{*}/k_{\mathrm{B}}T}, \qquad N_{\mathrm{nuc}} = Jt_{\mathrm{p}} \geq N_{\mathrm{req}}$$
$$\iff \Delta G_{\mathrm{het}}^{*} \leq B_{\mathrm{acc}} = k_{\mathrm{B}}T\ln(J_0 t_{\mathrm{p}}/N_{\mathrm{req}}) \tag{13}$$

7 Conformality (C4, R-slurry)

A barrier needs a continuous film; whether the mineral spreads or beads is a wetting balance, not a barrier height.

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}] \tag{14}$$

$S_{\mathrm{spread}} \geq 0$: no finite θ ($\theta \to 0$, $f \to 0$), layer-by-layer growth. $S_{\mathrm{spread}} < 0$: $\theta > 0$, island growth; a film forms only on coalescence, at cap height $h_{\mathrm{c}} = \tan(\theta/2)/(2\sqrt{N_{\mathrm{s}}})$, and closes at $h_{\mathrm{cl}} = \sqrt{2}h_{\mathrm{c}} \leq d_{\max}$ iff $N_{\mathrm{s}} \geq \tan^2(\theta/2)/(2d_{\max}^2)$ [m; m^{-2}]. A small-θ bound and $S_{\mathrm{spread}} \geq 0$ are one criterion at different strictness ($S_{\mathrm{spread}} = \gamma_{\mathrm{nl}}(\cos\theta - 1)$); coalescence is the independent one. A low f does not imply coverage: $f(\pi/2) = \frac{1}{2}$ is still island growth.

8 Stability (C5): screens, necessary not sufficient

(a) R-slurry, aqueous Pourbaix, per non-O, non-H atom [$\mathrm{J\,mol^{-1}}$]:

$$\Psi_\phi = \tfrac{1}{n_{\mathrm{M}}}\big[G_\phi - n_{\mathrm{O}}\mu_{\mathrm{H_2O}} - RT\ln 10\,(2n_{\mathrm{O}} - n_{\mathrm{H}})\,\mathrm{pH} - (2n_{\mathrm{O}} - n_{\mathrm{H}} + Q)FE_{\mathrm{SHE}}\big] \quad (15)$$

$\Delta G_{\mathrm{pbx}} = \Psi_\phi - \Psi_{\mathrm{hull}} \geq 0$; zero means stable at (pH, E_{SHE}).

(b) R-cell, Li open, E_{Li} vs $\mathrm{Li^+/Li}$ [$\mathrm{J\,mol^{-1}}$; V]:

$$\mu_{\mathrm{Li}} = \mu^\circ_{\mathrm{Li}} - FE_{\mathrm{Li}}, \qquad \Phi_\phi = G_\phi - n_{\mathrm{Li}}\mu_{\mathrm{Li}} \quad (16)$$

Window $[E_{\mathrm{red}}, E_{\mathrm{ox}}]$: the E_{Li} over which ϕ lies on the Li-open hull.

9 Insulating film (C4 limitation, R-cell)

The mineral is an electronic insulator, so conformality cannot be bought with thickness; the film must still pass the collector current.

$$R_A = \rho_{\mathrm{e}} d \leq R_{A,\max} \;\Rightarrow\; d \leq R_{A,\max}/\rho_{\mathrm{e}} \quad [\Omega\,\mathrm{m^2};\ \mathrm{m}] \quad (17)$$

$$j = j_0 \mathrm{e}^{-2\kappa d}, \quad \kappa = \frac{\sqrt{2m_{\mathrm{e}}\phi_{\mathrm{B}}}}{\hbar} \;\Rightarrow\; d \leq \frac{\ln(j_0/j_{\mathrm{req}})}{2\kappa} \quad [\mathrm{m}] \quad (18)$$

Symbols

generated from `theory/symbols.yaml`; Reg.: S = R-slurry, C = R-cell; – = dimensionless

Symbol	Meaning	SI unit	Reg.
Exact SI constants (kept symbolic)			
k_B	Boltzmann constant	$\mathrm{J\,K^{-1}}$	S,C
N_A	Avogadro constant	$\mathrm{mol^{-1}}$	S,C
R	molar gas constant	$\mathrm{J\,K^{-1}\,mol^{-1}}$	S,C
F	Faraday constant	$\mathrm{C\,mol^{-1}}$	S,C
e	elementary charge	C	S,C
$\hbar$	reduced Planck constant	J s	C
m_e	electron mass	kg	C
State variables			
T	absolute temperature of the regime	K	S,C
pH	pH (molal scale)	–	S
E_SHE	collector potential vs SHE	V	S
E_Li	collector potential vs $\mathrm{Li^+/Li}$	V	C
Adhesion in water (C1)			
γ_ow	oxide-water interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_pw	polymer-water interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_op	oxide-polymer interfacial free energy in contact	$\mathrm{J\,m^{-2}}$	S
γ_o	oxide surface free energy against vacuum	$\mathrm{J\,m^{-2}}$	S
γ_p	polymer surface free energy against vacuum	$\mathrm{J\,m^{-2}}$	S
γ_w	water surface free energy against its vapour	$\mathrm{J\,m^{-2}}$	S
$W_\mathrm{adh,wet}$	work of adhesion in water (Dupre)	$\mathrm{J\,m^{-2}}$	S
$W_\mathrm{adh,dry}$	work of adhesion against vacuum (comparator)	$\mathrm{J\,m^{-2}}$	S
ΔF_PMF	PMF free energy, separated minus bound	J	S
A_c	fragment contact area	$\mathrm{m^2}$	S
Sequestration (C2)			
ΔG_seq	$\mathrm{Ca^{2+}}$ transfer free energy, bulk to binding layer	$\mathrm{J\,mol^{-1}}$	S
ΔG_site	transfer free energy to the deprotonated site	$\mathrm{J\,mol^{-1}}$	S
$\mathrm{p}K_\mathrm{a}$	pKa of the coordinating group	–	S
$\mathcal{E}$	$\mathrm{Ca^{2+}}$ enrichment factor (dilute limit)	–	S
c_bulk	bulk $\mathrm{Ca^{2+}}$ concentration	$\mathrm{mol\,m^{-3}}$	S
c_local	mean $\mathrm{Ca^{2+}}$ concentration in the layer	$\mathrm{mol\,m^{-3}}$	S
c_max	saturation concentration of the layer	$\mathrm{mol\,m^{-3}}$	S
Γ_s	areal binding-site density	$\mathrm{mol\,m^{-2}}$	S
δ_s	binding-layer thickness	m	S
V_site	volume per binding site	$\mathrm{m^3}$	S
x	Langmuir binding variable	–	S
Θ	fraction of sites occupied by $\mathrm{Ca^{2+}}$	–	S
Γ_Ca	$\mathrm{Ca^{2+}}$ surface excess (Gibbs)	$\mathrm{mol\,m^{-2}}$	S
Supersaturation			
$a_\mathrm{Ca2+}$	bulk $\mathrm{Ca^{2+}}$ activity	–	S
$a_{\mathrm{CO_3^{2-}}}$	bulk carbonate activity	–	S
K_sp	solubility product of the registered framing	–	S
Ω_bulk	bulk saturation ratio IAP/Ksp	–	S
Ω	saturation ratio where the nucleus forms	–	S
$\Delta\mu$	driving force per formula unit	J	S
Nucleation and Young (C3)			
γ_nl	nucleus-liquid interfacial free energy	$\mathrm{J\,m^{-2}}$	S
γ_sl	template-liquid energy at the regime activities	$\mathrm{J\,m^{-2}}$	S
$\gamma_\mathrm{sl}^{(0)}$	template-liquid energy without $\mathrm{Ca^{2+}}$ binding	$\mathrm{J\,m^{-2}}$	S
$\Delta\gamma_\mathrm{ads}$	$\mathrm{Ca^{2+}}$ adsorption change of the template-liquid energy	$\mathrm{J\,m^{-2}}$	S
γ_sn	template-nucleus energy in water (= ref + correction)	$\mathrm{J\,m^{-2}}$	S
$\gamma_\mathrm{sn}^{(0)}$	template-nucleus energy, reference without water displacement	$\mathrm{J\,m^{-2}}$	S
$\Delta\gamma_\mathrm{w}$	water-displacement correction (cost > 0)	$\mathrm{J\,m^{-2}}$	S
γ_eff	deck effective nucleus-liquid energy (audit only)	$\mathrm{J\,m^{-2}}$	S
θ	contact angle, measured through the nucleus	rad	S
$f(\theta)$	heterogeneous shape factor	–	S
V_m	molar volume of the registered framing	$\mathrm{m^3\,mol^{-1}}$	S
r^*	critical radius	m	S
ΔG^*_hom	homogeneous nucleation barrier per nucleus	J	S
ΔG^*_het	heterogeneous barrier on the template	J	S
Accessibility of the barrier (C3)			
J_0	heterogeneous nucleation prefactor per area	$\mathrm{m^{-2}\,s^{-1}}$	S
t_p	time available for nucleation	s	S
N_req	required areal density of nuclei	$\mathrm{m^{-2}}$	S
B_acc	largest accessible barrier	J	S
N_nuc	expected areal density of nuclei	$\mathrm{m^{-2}}$	S
Conformality (C4)			
S_spread	spreading coefficient	$\mathrm{J\,m^{-2}}$	S
N_s	island density at coalescence	$\mathrm{m^{-2}}$	S
h_c	geometric coalescence thickness	m	S
h_cl	geometric closure thickness	m	S
Stability (C5)			
G_ϕ	molar Gibbs energy of species phi	$\mathrm{J\,mol^{-1}}$	S
$\mu_{\mathrm{H_2O}}$	chemical potential of water	$\mathrm{J\,mol^{-1}}$	S
n_O	number of O atoms per formula unit	–	S
n_H	number of H atoms per formula unit	–	S
n_M	number of non-O, non-H atoms	–	S
Q	species charge in units of e	–	S
Ψ_ϕ	Pourbaix potential per metal atom	$\mathrm{J\,mol^{-1}}$	S
Ψ_hull	lower envelope of Psi (the hull)	$\mathrm{J\,mol^{-1}}$	S
ΔG_pbx	energy above the aqueous Pourbaix hull	$\mathrm{J\,mol^{-1}}$	S
n_Li	number of Li atoms per formula unit	–	C
μ°_Li	Li chemical potential in Li metal	$\mathrm{J\,mol^{-1}}$	C
μ_Li	Li chemical potential at the electrode potential	$\mathrm{J\,mol^{-1}}$	C
Φ_ϕ	Li-open grand potential	$\mathrm{J\,mol^{-1}}$	C
ΔG_r	reaction free energy against Li metal	$\mathrm{J\,mol^{-1}}$	C
Δn	Li released by the reaction	–	C
$\Delta\Phi$	grand-potential change of the reaction	$\mathrm{J\,mol^{-1}}$	C
E^*	potential limit set by one reaction	V	C
E_red	lower limit of the stability window	V	C
E_ox	upper limit of the stability window	V	C
Insulating film (C4, R-cell)			
d	film thickness	m	C
ρ_e	film electronic resistivity	$\Omega\,\mathrm{m}$	C
R_A	area-specific film resistance	$\Omega\,\mathrm{m^2}$	C
$R_{A,\mathrm{max}}$	admissible area-specific resistance (input)	$\Omega\,\mathrm{m^2}$	C
ϕ_B	tunnelling barrier height	J	C
κ	tunnelling decay constant	$\mathrm{m^{-1}}$	C
j_0	tunnelling prefactor current density	$\mathrm{A\,m^{-2}}$	C
j_req	required current density (input)	$\mathrm{A\,m^{-2}}$	C
$d^\mathrm{tun}_\mathrm{max}$	thickness bound, direct tunnelling	m	C
d_max	admissible film thickness	m	C
$d^\mathrm{ohm}_\mathrm{max}$	thickness bound, ohmic limit	m	C
Registration constants (theory/thresholds.md)			
p	credible level of the decision rule	–	S,C
p_deg	least mass shift that counts as degradation	–	S,C
m_W	C1 margin on the wet work of adhesion	$\mathrm{J\,m^{-2}}$	S
m_seq	C2 margin on the $\mathrm{Ca^{2+}}$ transfer free energy	$\mathrm{J\,mol^{-1}}$	S
m_cos	C3 margin on cos theta	–	S
m_S	C4 margin on the spreading coefficient	$\mathrm{J\,m^{-2}}$	S
e_f	expected error of a computed formation energy	$\mathrm{J\,mol^{-1}}$	S,C
n_at	atoms per formula unit	–	S,C
Δ_ACC	ACC offset above calcite	$\mathrm{J\,mol^{-1}}$	S
τ_hull	C5 hull tolerance	$\mathrm{J\,mol^{-1}}$	S
m_E	C5 margin on the window limits	V	C
$E^\mathrm{min}_\mathrm{cell}$	lower end of the R-cell potential range	V	C
$E^\mathrm{max}_\mathrm{cell}$	upper end of the R-cell potential range	V	C

Conventions. Angles in rad; free energies per formula unit (J) or per mole ($\mathrm{J\,mol^{-1}}$) as stated; activities on the molal scale; potentials vs SHE in R-slurry and vs $\mathrm{Li^+/Li}$ in R-cell. Ω is the only supersaturation variable (the charter's S for C3 is Ω). The functions W4 implements, with exact limits, monotonicity and the sign-flip tests, are in `theory/contract.yaml`.